

Lab 4 Report

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Page 1

1. The Arduino isn't capable of storing much data long-term by itself because it usually stores data in volatile memory. What is the difference between volatile and non-volatile memory? Look at the online Arduino documentation to answer this. Provide a link to where you found this information as well as a screenshot of the webpage.

ANS) Volatile memory is erased when the power is cut. Non-volatile is able to be retained after the power is cut.

<https://docs.arduino.cc/learn/programming/memory-guide/>

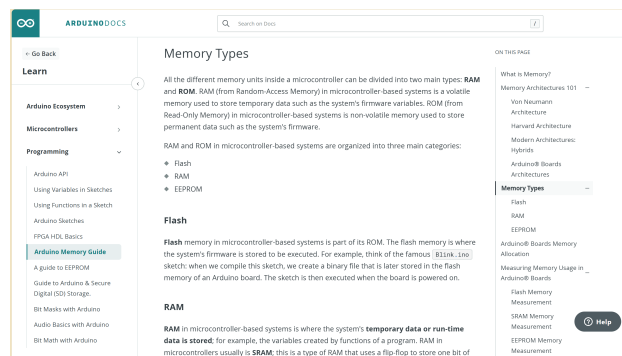


Figure 1: memory

2. Which type of memory stores Arduino sketches (e.g. like Blink.ino)?

ANS) Sketches are sent to flash

3. 1. Memory

- a. How much non-volatile memory does the Arduino have?

Flash Memory: 32 KB (stores the program code).

EEPROM: 1 KB (stores small amounts of data that persist after power-off).

- b. Find one product for Arduinos to store data longer-term and provide a link.

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For long-term storage of large amounts of data, a MicroSD Card Module is typically used.

Product: Adafruit MicroSD card breakout board+

Link: <https://www.adafruit.com/product/254>

2. Pedometers

- a. Pedometers are often placed on the hip or wrist. Based on the results of the lab, what pros and cons do you think each of these positions would have?

Hip Placement: Better for accuracy as it measures the vertical movement of the center of mass, but less convenient for the user to interact with.

Wrist Placement: Highly convenient for viewing data, but prone to false counts from arm gestures (reaching, typing, or moving hands while talking).

- b. Try moving the tilt ball switch around in different patterns (e.g. moving your hand as if you were moving a mouse). Does your pedometer count more or fewer steps while doing these activities? Why do you think this might be occurring?

It typically counts **more** steps. This is because a tilt ball switch is a mechanical sensor; any vibration or irregular movement causes the ball to rattle against the internal contacts. Without software filtering (debouncing), the Arduino interprets every small mechanical bounce as a new step.

Page 2

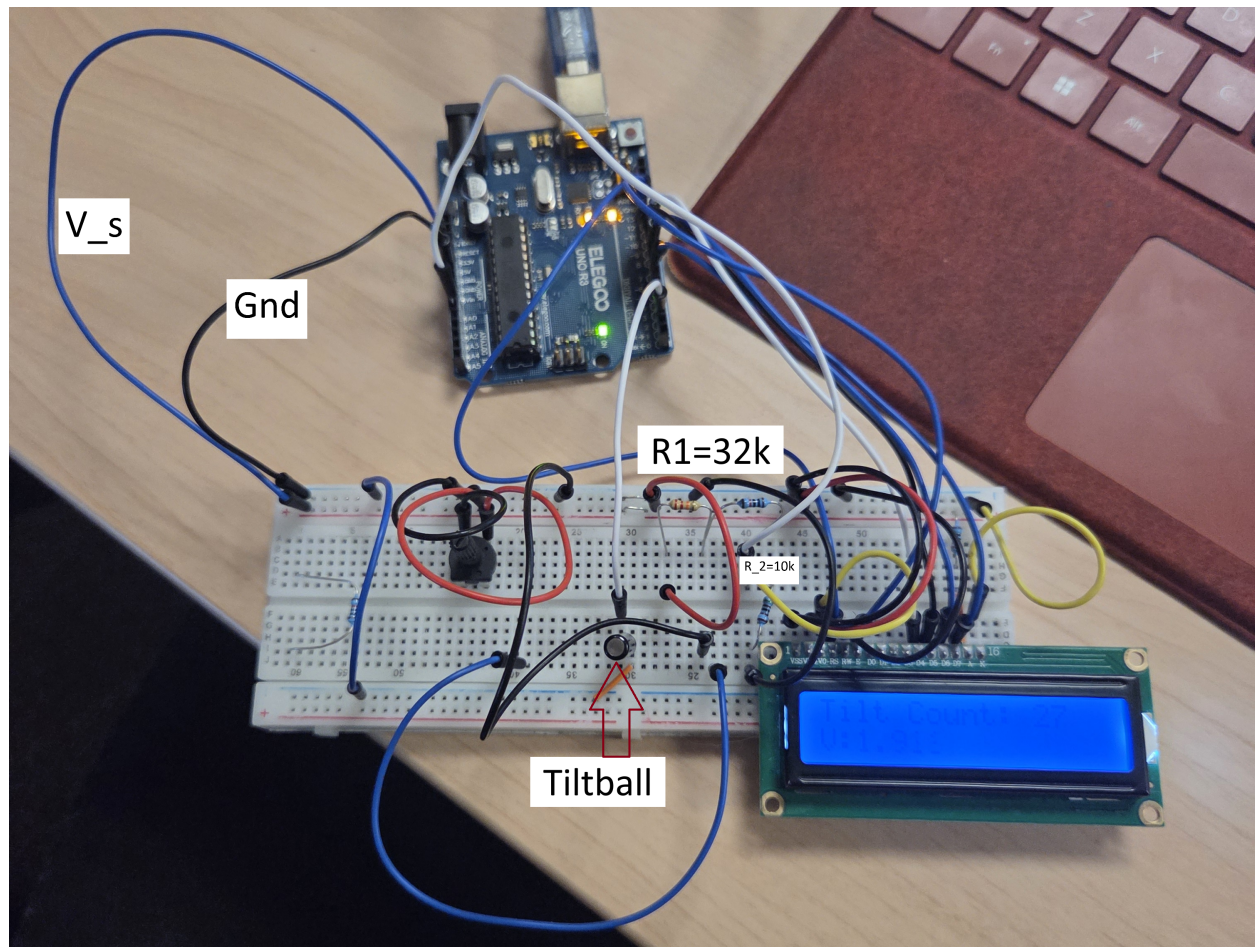
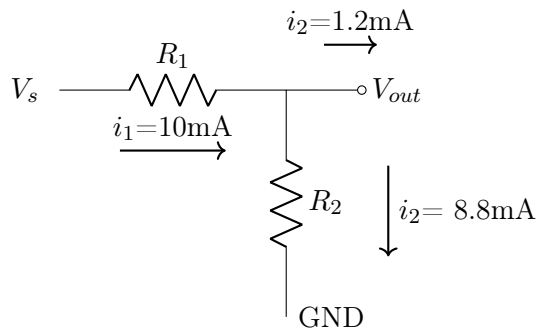
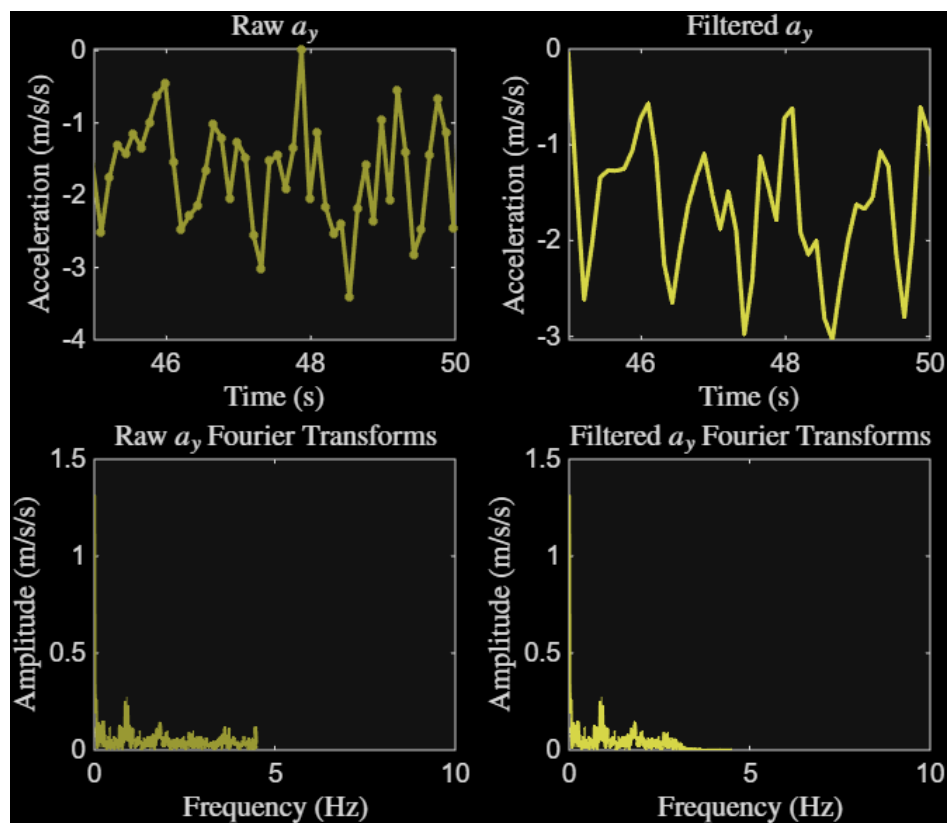
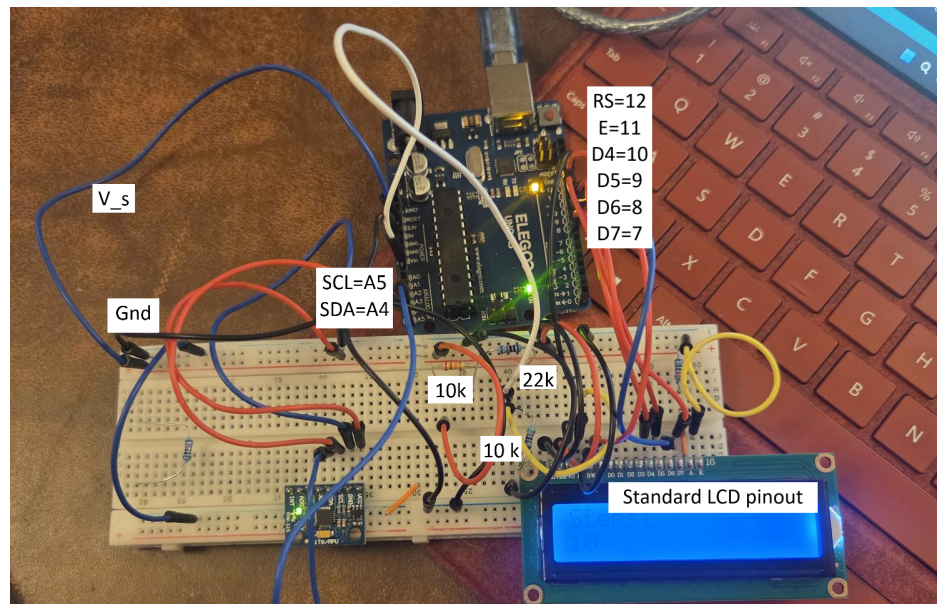


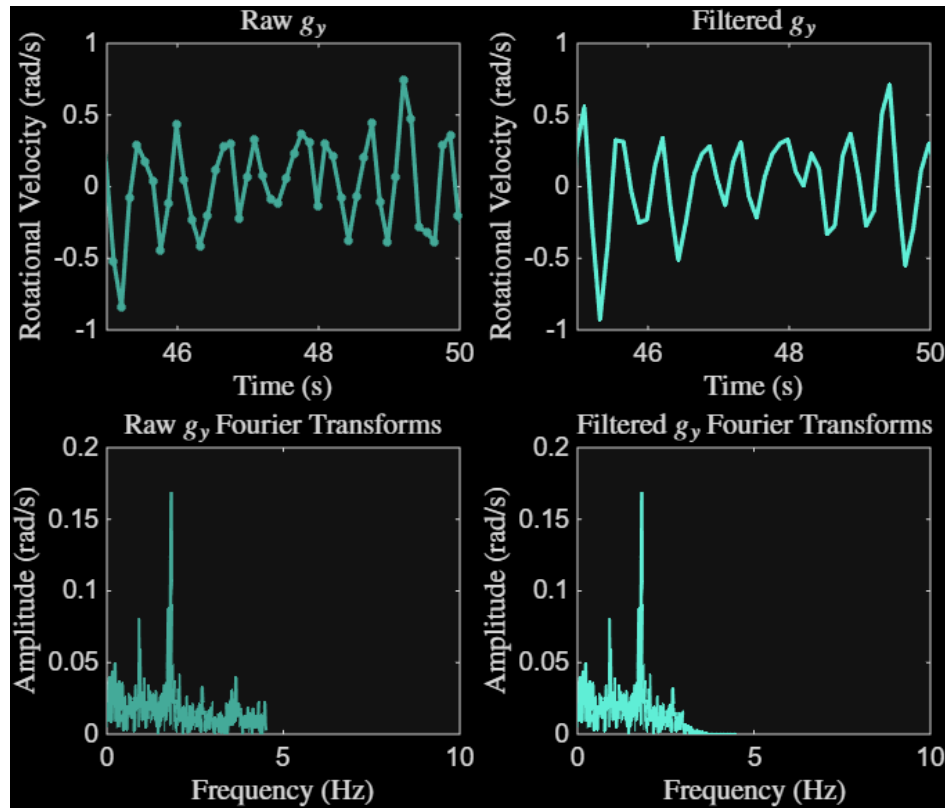
Figure 2: board

The pros of a tilt switch are that it can be very good at showing when the slightest moment has happened. Cons would be that they are soooo noisy. From the tech manuals I found that we have a (i_3) 1.2 mA draw for the LCD and a supply of (i_1) 10 mA from the arduino. This would leave (i_3) 8.8 mA going over the bottom resistor of the voltage divider. I wasn't able to find a source that could give me a decent answer to the lifespan. I did find some website that mentioned something about the LCD acting in a capacitive nature but I wasn't sure what that meant. With the 1.2mA draw though I found that it would run for about 19 Days on the standard 9V battery.



Topic: Page 3





I was spending too much time figuring out how to sterilize the board and hardware (juggling a bread board plugged into the arduino by flimsy wires was a pain.). In the end i just took a data set from a walk around campus and just found the normalized data plot and just used the values at $-1\sigma + .9$. I just kept my f_{cutoff} from the last lab and ran with the butters filter hoping that the high freq noise would be killed well enough..

4. **Page 4:** List online sources and collaborators.

no additional online sources outside of the lab links. No other students helped.

5. Page 5: Self-evaluation:

- (a) What did you expect to learn? I hoped to learn about being able to use Matlab to do data visualization
- (b) Rate class participation and needed adjustments. I feel like I did a great job staying on top of the assignments.
- (c) Grade yourself (0–4) and explain. I would give myself a 4. I had the majority of the lab done by tuesday of this week.
- (d) What did you actually learn? Getting into Matlab was actually nice. I feel bad about my pervious hesitation but its obviously a better tool for transformations and general "mathy" physics stuff.

Code

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%CODE HERE
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